

Comparing Robots and Non-robot Phygital Artefacts in Children's Storytelling via a Systematic Review

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Abstract

Storytelling is a widely explored educational practice. Within Human-Robot Interaction (HRI), robots are extensively employed for storytelling. In parallel, non-robot phygital artefacts — physical objects augmented with digital technologies — have been considered for supporting interaction and collaboration. However, systematic comparisons between these approaches remain limited. This paper presents a systematic review of 135 out of 1,040 studies (2014–2025) involving participants under 18. The review compares robots and non-robot phygital artefacts used in storytelling. Studies were coded across analytic lenses on media, participation, collaboration, learning goals, and Artificial Intelligent support, enabling age-stratified comparison. Findings reveal that non-robot phygital artefacts more consistently foster peer collaboration and creative ownership, while robots primarily contribute social presence and novelty, often structuring participation sequentially. The review highlights implications for future HRI design, suggesting that robots should complement alternative phygital artefacts by promoting openness, adaptability, and collaboration in storytelling.

CCS Concepts

• Computer systems organization → Robotics; • Applied computing → Interactive learning environments.

Keywords

Storytelling, Children, Robots, Phygital Artefacts, Collaboration, STEM, Artificial Intelligence, Systematic Literature Review

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1 Introduction

Storytelling is a social and cultural practice centred on the narration and sharing of stories, widely acknowledged as a powerful pedagogical approach, particularly in childhood education [25, 26, 42, 60]. Prior research has shown that storytelling enhances literacy and writing skills [101, 153], fosters collaborative and cooperative learning by encouraging the co-construction of knowledge [59, 74], and improves overall engagement [130].

With the spread of digital technologies, storytelling is today often mediated by devices such as tablets, smartphones, and laptops. As a result, educational digital storytelling (DS) has been widely investigated [35, 41, 173] and has become a well-established practice in school contexts [3, 130]. While these approaches have expanded access to narrative creation, they have also reinforced screen-centred interaction. More recently, storytelling research has explored alternatives that move beyond screens towards *phygital artefacts*, namely physical-digital systems combining tangible manipulation with computational capabilities [57, 62, 103]. In the literature, such systems are referred to using different labels, including *embedded*, *augmented*, or *hybrid* artefacts [99, 102]. In this paper, we use phygital artefacts as an umbrella term for these systems.

In storytelling, non-robot phygital artefacts typically behave as shared resources that children can collectively manipulate to construct narratives. By foregrounding tangible interaction and material engagement, these artefacts support open-ended exploration, reduce barriers to participation, and foster collaborative narrative construction and meaning-making [43, 58, 129, 141].

Alongside phygital artefacts, in Human-Robot Interaction (HRI), robots are frequently proposed as compelling media for narrative activities due to their embodiment and social presence. Robots offer new ways for children to engage with stories, whether acting as storytellers, listeners, or co-creators [110, 111].

These approaches broaden the spectrum of tools available for storytelling and narrative learning, but they also raise questions about how robots differ from other phygital artefacts in shaping children's agency and collaboration. Despite their potential, research remains fragmented. Existing studies tend to focus either on phygital toolkits [57] or on robot-mediated storytelling [110, 111], and rarely offer a systematic comparison across modalities or age groups. Understanding where the strengths of robots and non-robot phygital artefacts converge or diverge is crucial to inform future HRI design for children's agency and collaboration.

Beyond its educational applications, storytelling is increasingly shaped by automation, screen-dominated interaction, and Artificial Intelligence (AI)-driven content generation, which risk marginalising shared authorship and social interaction in storytelling practices. In this context, robots act as socially embodied partners that actively shape interaction, participation, and collaboration in storytelling activities. A systematic comparison with non-robot phygital artefacts is therefore needed to clarify how different phygital technologies distribute collaboration and ownership in children's storytelling.

To address this gap, we present a systematic literature review of 135 studies, published between 2014 and 2025, on storytelling technologies for participants under 18. The review focuses on robots and non-robot phygital artefacts, excluding purely digital systems without a physical component. The analysis contrasts these two modalities across age groups and five lenses adapted from prior literature: media use, children's role, collaboration, educational goals, and AI support [67, 159]. By systematically comparing these modalities, the paper maps the state of the art and highlights design implications for future HRI, showing how robots might build on the strengths of alternative phygital artefacts, particularly in fostering collaboration and agency, rather than replacing them.

2 Related Work

Several reviews map the educational benefits of DS, outlining design patterns and curricular applications across child age groups [35, 161, 173]. While these mappings inform our rationale, they generally focus on screen-based platforms and lack systematic attention to *phygital artefacts* or *robot embodiment*. Accordingly, we delimit the scope of our counting to artefacts where a tangible, phygital, or mixed-reality component is integral to interaction.

2.1 Robots for storytelling

Within HRI and educational robotics, robots have been explored as mediators of storytelling activities with children. Recent systematic reviews highlight positive impacts on teaching and learning, yet note that robots most often *present* stories rather than *co-author* narratives with children [52]. Age stratification is rarely explicit, and collaboration or agency lenses are seldom applied.

Closest to our scope are the studies of Palioura et al. [110, 111], who conducted scoping reviews of educational robotics combined with storytelling. Their analyses consider multiple aspects, including research settings, types of robots, specific children's role in storytelling (e.g., active, passive), and educational goals. These works establish a valuable lens-based approach that aligns with our own framing. However, important differences remain: (i) their scope is limited to *robotic* solutions, without considering non-robot phygital artefacts; (ii) their age ranges (4–22 years) are not treated as non-overlapping denominators, limiting developmental insights; and (iii) their lenses, while similar (e.g., role of children), do not systematically cover *screen use* and *AI support*.

More broadly, reviews on robots in classrooms or special education emphasise learning outcomes and affective gains [83, 108, 113, 170, 172], but they seldom analyse the co-creation processes central to storytelling. Overall, robots point to clear pedagogical potential but leave gaps regarding collaboration, and direct comparison with non-robotic phygital artefacts.

2.2 Non-robot phygital artefacts for storytelling

Phygital, tangible, and mixed-reality artefacts have been widely explored to support children's storytelling from early childhood onwards. Reviews of tangible user interfaces and phygital toolkits consistently report benefits for inclusion, cooperation, and engagement, showing how hands-on interaction can scaffold the co-construction of narratives [118, 126, 138]. The review by Gennari et al. further consolidates this evidence by mapping design goals and interaction modalities across technology-enhanced storytelling toolkits [57].

Complementing these storytelling-focused surveys, Torres et al., provides a broader perspective with a systematic review of physical-digital play technologies for children aged 0–12 [159]. Their work identifies four developmentally relevant behaviours fostered by phygital play: self-monitoring, collaboration, decision making, and problem solving/physical activity and highlights design mechanisms that afford these behaviours. Although not centred on storytelling, these findings strengthen the rationale for examining collaboration and agency in non-robotic phygital artefacts.

Taken together, this literature demonstrates the potential of tangible and phygital approaches to enhance children's agency and multimodal expression. Yet prior reviews rarely stratify findings by *age range*, lack explicit denominators for quantitative comparison, and, most critically, do not contrast non-robot phygital systems with robots ones. Addressing these gaps, our review provides an age-stratified, lens-wise synthesis that systematically compares robots and non-robot phygital artefacts for children's storytelling.

3 Methodology

This research conducts a systematic literature review following Kitchenham's [78] and PRISMA [109] guidelines.

The process was divided into 3 stages: (i) *Planning*, defining the objectives and the Research Questions (RQs); (ii) *Conducting*, including the selection of digital libraries, formulation of search strings, and definition of inclusion and exclusion criteria; and (iii) *Reporting*, where results were analysed through lenses derived from prior literature [67, 159].

3.1 Scope and definitions

This review focuses on interactive technologies that enable learners up to 18 years old to engage in storytelling via any phygital device.

Articles are classified into two categories:

Robots. Following ISO 8373:2021 [1], a robot is an actuated mechanism with a degree of autonomy that moves within its environment to perform intended tasks. Consistent with HRI scholarship [63], we operationalise a robot in our coding as a physically embodied system (three-dimensional body) capable of acting upon its environment with some level of autonomous behaviour, irrespective of degrees of freedom. Robots may act as storytellers, listeners, or co-creators. Purely screen-based agents and teleoperated devices without embodied autonomy (e.g., a tablet on wheels) are excluded.

Non-robot phygital artefacts. Non-robot phygital artefacts integrate physical components with digital interactivity but *do not* include an autonomous embodied agent. Examples include tangible storytelling kits, IoT-enabled objects, sensor-augmented wearables, interactive tabletops, and mixed-reality environments [159].

The review is guided by two Research Questions (RQs):

- RQ1 How are storytelling practices distributed across robots and non-robot phygital artefacts, and how do these distributions vary by age range?
- RQ2 How do the robot and non-robot phygital artefact differ across key storytelling lenses, including media use, children's role, collaboration, educational goals, and AI support?

3.2 Inclusion & Exclusion Criteria

The screening was jointly discussed by four researchers. A study was retained as *relevant* if it satisfied all inclusion (I) and none of the exclusion (E) criteria:

- I1 Published between Jan. 2014 and Feb. 2025
- I2 Fully written in English, beyond just the abstract.
- I3 Peer-reviewed venue.
- I4 Reports on artefacts and devices enhanced with interactive technologies for storytelling.
- I5 Target users under 18.
- E1 Duplicate or extension of an included paper.
- E2 Purely digital storytelling (no tangible device).
- E3 Target users not specified and not inferable.
- E4 Two-page idea papers only.

3.3 Data Sources and Database Search

To construct the search string, keywords were extracted from the RQs, and alternative terms were considered. The search string was formulated by combining keywords using conjunctions (AND) and alternative terms using disjunctions (OR). Four researchers independently analysed search queries for online databases. They revised queries and discussed outcomes iteratively. The final query was used to search for papers with those terms occurring in the title, abstract, and keywords:

(storytelling OR "story telling" OR story-telling) AND (tangible OR phygital OR physical OR smart OR electronic* OR artefact* OR artifact* OR "internet of thing*" OR iot OR toy* OR robot* OR device*) AND (child* OR scholar* OR learner* OR student* OR pupil* OR kid*)*

This review targets peer-reviewed contributions indexed in the Scopus and ACM Digital Library databases. The search on Scopus was restricted to Computer Science, Engineering, Arts and Humanities to find relevant papers suitable for our study. A total of 1,040 records (799 from Scopus and 241 from ACM Digital Library) matched the search string and are compliant with I1 and I2.

3.4 Study Selection and Screening

The complete selection process is illustrated in Figure 1, following the PRISMA 2020 framework [109]. Duplicates were removed with Zotero, yielding 885 records. Five additional papers were identified through expert input and hand searches major journals and conferences related to human computer interaction, child computer interaction, and education, and in related surveys. In total, 890 non-duplicated papers were identified.

All records were imported into a spreadsheet for coding. Two reviewers independently screened titles and abstracts, selecting 222 papers for full-text review. Disagreements ($n = 71$) were resolved by

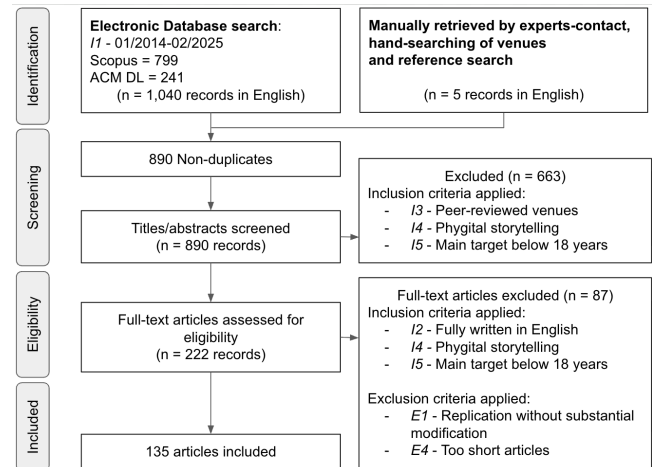


Figure 1: The PRISMA-based review methodology.

discussion. Each researcher then cross-checked 30% of the other's coding, leading to the reclassification of 24 studies as irrelevant, mainly for not meeting the target age (I5) or focusing only on digital artefacts (I4). A further 34 irrelevant papers were jointly discussed, with one reclassified as relevant.

During full-text review, ten articles were excluded according to E4. For thirteen paywalled articles, authors were contacted; five provided access. In cases of duplicate reports on the same toolkit, the most recent or extensive version was retained, preferring journal publications, which led to the exclusion of 25 papers. Five further studies were removed because the target age was not explicitly stated, while generic mentions of *children* or *young learners* were accepted. An additional 18 were deemed off-topic, e.g., focusing only on character design, recording tools, or ambient effects without interactive storytelling. In total, 135 studies were included.

3.5 Reporting and Categorization of Outcomes

To analyse the selected corpus, we coded each study using a set of analytic lenses (L1–L5) adapted from prior reviews on storytelling, educational robotics, and phygital play [67, 111, 159]. These lenses capture dimensions of agency, collaboration, and automation relevant to HRI design for children.

- L1. Media.** Whether storytelling was mediated by screens (using screens as input or output device) or supported through screenless interaction. Screen-based mediation can constrain embodied engagement, while screenless artefacts promote head-up participation and inclusivity [43, 58, 67].
- L2. Role.** Whether children contributed actively (authoring or enacting stories) or passively (listening only) [67, 111]. Active roles were further specified (e.g., programmer, scriptwriter, performer), but these subcodes were collapsed into a binary active/passive distinction for synthesis. Agency is central to HRI because it determines whether robots and phygital artefacts empower children as creators or position them as audience.
- L3. Collaboration.** Following Roschelle and Teasley [127], we define collaboration as coordinated, synchronous activity aimed at maintaining a shared conception of the story, as opposed

to looser cooperation based on task division. In our coding, collaboration refers to real-time co-construction, while cooperation includes sequential or divided contributions mediated by robots or tools. While following the framework [127], we acknowledge that sequential or turn-based participation, particularly in robot-mediated settings, may still scaffold collaborative processes by supporting shared understanding, even if coded as cooperation. This lens is relevant to HRI because artefacts scaffold group dynamics and ownership.

- L4. STEM.** Whether studies explicitly targeted STEM (Science, Technology, Engineering, Mathematics) objectives, as opposed to literacy, creativity, or socio-emotional aims. STEM was included due to its prominence in global education agendas and in educational robotics research [57, 108].
- L5. AI support.** This lens examines the presence and type of AI used to guide the storytelling process [67, 159]. We distinguish three levels: (i) automation or rule-based systems (e.g., pre-programmed scripts); (ii) adaptive or algorithmic decision support, where the system adjusts feedback or narrative elements to user input; and (iii) generative or co-adaptive AI, which dynamically creates or modifies content in response to children's actions.

Coding was performed iteratively by two pairs of reviewers for each lens. The first reviewer coded all articles, recording the relevant excerpts supporting each coding decision. The second reviewer cross-checked all codes and excerpts, resolving disagreements through discussion. It resulted in five articles moved across categories. Pairs of reviewers are different for each lens. When information for a given lens was missing or ambiguous, it was coded as *Unclear*.

Studies were analysed by category (robots vs. non-robot phygital artefacts) and split by age range, since developmental and educational stages strongly affect how learners engage with storytelling technologies. Children, adolescents, and young adults differ in cognitive abilities, interaction styles, and learning needs. Since age typically aligns with schooling stages and curricular expectations, splitting results by age allows for a more systematic comparison across studies. A co-occurrence analysis of coded variables across the included studies allowed us to identify patterns across individual lenses and the age used as a clustering variable. This perspective also highlights design implications, helping to identify which technologies, methods, or interventions are more suitable for specific learner groups and ensuring that the synthesis reflects both theoretical and practical relevance.

3.6 Quality criteria, Biases and Limitations

To reduce the risk of low-quality evidence, we restricted the review to peer-reviewed studies and excluded very short papers (fewer than two pages), which provide insufficient methodological detail. Reliability was strengthened through a multi-stage screening process and the involvement of four independent annotators.

All data were extracted manually by reviewers, which guaranteed contextual accuracy but may have introduced subjective interpretation. The review was limited to Scopus and the ACM Digital Library, potentially omitting relevant contributions. Nevertheless, these two sources provide broad coverage of research in children, interaction, and education, mitigating the risk of missing

a substantial body of work. In addition, the automatic extraction procedure was complemented by a manual, expert-driven search.

Finally, to refine scope, the search deliberately excluded the keyword “digital”, thus omitting studies on purely screen-based storytelling. While this may have filtered out certain hybrid cases, the large corpus retrieved and the explicit inclusion criteria suggest that relevant studies were not systematically overlooked.

4 Results of the Literature Review

This section presents the findings of our review, based on 135 included studies. A slight majority of these, 71 articles (53%), employed robots, while the remaining 64 (47%) used non-robot phygital artefacts. Table 1 reports the distribution of studies by technology type and age group, distinguishing preschool (0–6), primary school (6–11), early adolescence (11–13), later adolescence (13–18), and mixed-age samples. Each study was counted once, either within a specific age range or, when multiple bands were covered, in a separate “mixed” category (avoiding overlaps with single ranges).

The following subsections unpack these distributions in greater detail. We first examine how storytelling artefacts are stratified by age (RQ1), and then compare robots and non-robot-based artefacts along the five analytic lenses (RQ2).

4.1 Age-stratified distribution (RQ1)

Most articles addressed mixed-age groups in both robot and non-robot studies (37% and 50% respectively). Table 1 summarises the 135 included studies by category (robot vs. non-robot) and age range (0–6, 6–11, 11–13, 13–18, Mixed).

Among studies that focused on a specific target group, the 6–11 age range was by far the most represented. Robots were employed in programming and creative authoring tasks (e.g., CreativeBot [50], Tangible Robotics for narratives [155]), while non-robot systems often involved tangible or mixed-reality toolkits for collaborative storytelling (e.g., STEAM City Kits [123], Multisensorial tangible user interfaces (TUIs) [139]).

The 0–6 preschool group formed the second largest cluster. Robots such as NAO and social robots were used as storytellers or listeners [55, 81], while non-robots included augmented books and tangible theatre [107, 147]. By contrast, adolescents were under-represented. Only a handful of studies targeted the 11–13 group, such as Gentoro robot for co-authoring [144] and the Asistranto tangible theatre [94]. For 13–18, robots like Pepper supported language learning [38], whereas only one non-robotic AR solution, StoryMakAR [61], addressed this group.

4.2 Lens-wise analysis (RQ2)

We coded all studies along five analytic lenses (L1–L5). The following paragraphs reports percentages and highlights illustrative studies.

Media (L1). Screen-based interaction emerged as a recurrent feature across age groups in both categories. In early childhood (0–6), most studies incorporated screens, with 47% of robot-based and 86% of non-robot-based studies relying on visual mediation. Robots such as NAO frequently combined verbal narration with tablet displays [81, 116], while phygital systems like *StoryCube* projected children's physical manipulations into digital narratives [169].

Table 1: Distribution of robots vs. non-robot phygital artefacts by age range and analytic lenses.

Age	Robot (71)							Non-robot (64)						
	#Art.	Media	Role	Collab.	STEM	AI	Ref.	#Art.	Media	Role	Collab.	STEM	AI	Ref.
0–6	17 (24%)	8	12	7	2	10	[20, 22, 26, 36, 44, 55, 71, 79, 81, 116, 117, 131, 142, 150–152, 163]	7 (11%)	6	7	5	0	4	[17, 28, 85, 107, 124, 147, 171]
6–11	20 (28%)	13	16	7	3	11	[11, 12, 18, 23, 39, 47, 50, 82, 86, 89, 93, 128, 133–136, 154–156, 175]	21 (33%)	14	19	16	8	11	[8, 13, 15, 34, 46, 54, 56, 64, 90, 91, 103, 104, 120–123, 139, 143, 162, 164, 166]
11–13	3 (4%)	3	3	2	1	2	[19, 144, 165]	3 (5%)	3	2	1	1	1	[65, 75, 94]
13–18	5 (7%)	3	3	1	0	1	[29, 30, 38, 92, 174]	1 (2%)	1	1	1	0	0	[61]
Mixed	26 (37%)	15	21	15	6	18	[2, 5, 6, 9, 10, 14, 16, 31, 40, 48, 49, 51, 53, 68, 69, 73, 80, 87, 88, 100, 106, 119, 125, 132, 145, 158]	32 (50%)	15	30	22	9	21	[4, 7, 21, 24, 25, 27, 32, 33, 37, 45, 62, 66, 70, 72, 76, 77, 84, 95–98, 105, 112, 115, 137, 140, 146, 148, 160, 167–169]

At the primary school stage (6–11), reliance on screens increase further (65% robots, 67% non-robots), spanning examples from robot-mediated platforms such as *CreativeBot* [50] to multisensorial TUIs that visualised children’s authored stories [139].

Among adolescents (11–13), screen use was universal across the two modalities, either to mediate interaction [19, 65, 75, 165] or to render the story visually [94, 144]. In the 13–18 group, prevalence remained high, with 60% of robot-mediated storytelling using screens mainly for authoring narratives [30, 92?], and 100% of non-robot artefacts employing them for story programming [61].

Mixed-age studies showed extensive use of screens (58% robot, 47% non-robot). Robots often used displays to scaffold the interaction between children and the robot [6, 14], while in non-robot systems, screens often visualized data captured from the physical environment projecting them into the digital story space through tabletop interfaces, interactive books, or gesture recognition models [4, 7, 21, 24, 27, 62, 76, 95, 115, 146, 148, 160, 169]. An original example is provided by Song et al. [140], who combined handicraft artefacts with a desktop-based mixed-reality environment.

Participants’ role (L2). Active participation was consistently high across age groups for both modalities, but more frequent in non-robot artefacts. Unlike collaboration (L3), no significant age effect was observed. In early childhood (0–6), 71% of robot-based and 100% of non-robot studies involved children as active contributors, e.g., programming Bee-bot movements [55] or manipulating toy-based kits [107]. At 6–11, active roles were even higher (80% robots, 90% non-robots), with children scripting robot dialogues [134] or co-authoring narratives via tangible interfaces [56]. Active roles were common in 11–13 (100% robots, 67% non-robots) and 13–18 groups (60% robots, 100% non-robots), e.g., multimodal authoring with Pepper [30]. Mixed-age groups showed the strongest emphasis on participation (81% robots, 94% non-robots), with children taking active narrative roles in both robotic [68] and phygital [76] settings. While most studies fostered active roles, the depth of agency varied. In several robot-mediated studies, children’s participation was limited to constrained responses or turn-taking with the robot (e.g., [81, 116]), whereas non-robot systems more often enabled open-ended co-creation through tangible manipulation (e.g., [56]).

Collaboration (L3). Collaboration varied across age groups and modalities. In preschool (0–6), collaboration was reported in 41% of robot-mediated studies compared to 71% of non-robot studies.

Children worked together to author stories by programming robot movements, as Blue-Bot [131], or taking turns to code stories [149].

Among primary school learners (6–11), collaboration emerged much more strongly in non-robot artefacts (76%) than in robot systems (35%), marking the widest gap between the two approaches. While robot-mediated collaboration often involved joint programming of robots such as Ozobot [157], non-robot artefacts as the *Interactive Storyteller* [8] or *RoboTale* [90] support story co-creation.

For adolescents aged 11–13, collaboration was less frequent overall, appearing in 67% of robot-mediated and 33% of non-robot studies. Robot-mediated storytelling sometimes encouraged group co-creation of performances [19] or cooperative navigation of storytelling robots such as GENTORO [144]. By contrast, non-robot artefacts emphasize collective listening activities [75].

In the 13–18 group, collaboration was rare, reported in only one study per category. In the robots category, it results in peer assessment [92], whereas non-robot storytelling focused on the collaborative authoring of interactive stories via *StoryMakAR* [61].

Mixed-age studies, by contrast, showed higher collaboration rates, with 58% of robot-mediated and 69% of non-robot artefacts fostering shared storytelling. Within the robot category, collaboration could occur both with peers and with the robot itself, for instance, by casting robots and children together as story characters [14], or by negotiating group consensus on the preferred continuation of a robot-narrated story [100]. In non-robot approaches, projects as *Project Torino* engaged participants in collaborative storytelling within a physical programming environment [72].

STEM as a Learning objective (L4). The inclusion of STEM as an explicit learning objective was unevenly distributed across age groups. In 0–6, only 12% of robot studies pursued explicit STEM goals, typically via coding toys such as *Bee-bot* that linked narrative

Table 2: Robot vs. non-robot comparison across analytic lenses (percentages of “Yes”; χ^2 /Fisher tests; effect sizes ϕ or V).

Lens	Robot	Non-robot	p	Effect size
L1. Media	59%	61%	.972	$\phi = .01$ (none)
L2. Role	77%	92%	.034*	$V = .21$ (small–med)
L3. Collab.	45%	70%	.005**	$V = .24$ (medium)
L4. STEM	17%	28%	.174	$V = .16$ (small)
L5. AI	59%	66%	.551	$\phi = .04$ (none)

play to programming trajectories [55], while none of the non-robot systems in this age group addressed STEM explicitly

At the primary school stage (6–11), STEM appeared in 15% of robot-mediated and 38% of non-robot studies. Examples include robot-based studies where programming tasks were embedded into storytelling [89], and non-robot artefacts such as *STEAM City Kits* [123] or *STEAM figurines* [121], which integrated curricular science content into narrative practices.

Among adolescents (11–13), STEM objectives were present in 33% of robot- and non-robot-based studies. Robot-mediated storytelling such as *Gentoro* [144] and robot-based classroom activities [19] embedded programming and problem-solving tasks into narrative practices, while non-robot artefacts like *Asistranto* [94] and interactive tabletop approaches [65] linked collaborative storytelling to logical reasoning and scientific exploration.

By contrast, in the 13–18 range, none of the included studies explicitly pursued STEM goals, focusing instead on language learning, multimodal expression, or cultural awareness [30, 38, 61, 92].

Mixed-age groups displayed the highest concentration of STEM-oriented studies, with 23% of robot-mediated and 28% of non-robot systems embedding STEM. Typical examples include maker-oriented kits combining storytelling with engineering activities [10].

AI support (L5). The integration of AI features was frequent across categories limiting to rule-based systems and adaptive decision support, though with differences by age group.

In early childhood (0–6), 59% of robot and 57% of non-robot artefacts integrated some form of automation, usually for scaffolding basic interactions. For instance, *Bee-bot* robots used pre-programmed rules to provide immediate feedback during coding-based storytelling [55], while non-robot kits such as *Iyagi* employed simple speech-recognition models to trigger narrative responses [28].

At 6–11, AI appeared in over half of the systems. Robot-mediated tools such as *CreativeBot* embedded adaptive tutoring modules to adjust the narrative flow according to children's inputs [50], whereas non-robot systems like the multisensorial TUI automatically processed sensor data (e.g., motion or object placement) to update and advance the story without requiring manual control [56].

For adolescents (11–13), AI was more common in robots (67%) than non-robot artefacts (33%), with robots often functioning as conversational partners, adapting prompts or responses in real time, as in *Gentoro* [144]. By contrast, the few non-robot artefacts relied mainly on rule-based adaptation.

In the 13–18 group, AI features were rare (20% of robot systems, none of non-robots), usually limited to authoring assistance such as speech-to-text modules [30].

Mixed-age groups showed the highest prevalence (69% robots, 66% non-robots). Examples include robots dynamically adjusting storytelling complexity to group composition [6] and AI-enhanced phygital books that analysed children's inputs (voice or gestures) to provide adaptive story continuations [76].

4.3 Lens-wise comparative results

To complement the age-stratified distributions in Table 1, we compared robot and non-robot phygital artefacts across the five analytic lenses. Percentages reflect the proportion of studies coded as Yes. Studies coded as *Unclear* for a given lens were excluded from the

denominator when computing percentages and statistical tests. The aggregate comparison is reported in Table 2, while Figure 2 provides a visual synthesis across age ranges.

To assess differences, we applied Pearson's χ^2 tests; when expected cell counts were below 5, Fisher's exact test was used. A Holm–Bonferroni correction was applied across the five lenses. In addition to p -values, we report effect sizes using Cramér's V (or ϕ for 2×2 tables). No significant differences emerged for *Media*, *STEM goals*, or *AI support*, whereas *Active role* and *Collaboration* differed significantly between modalities (Table 2). At the age range level, the only statistically significant difference was observed for Collaboration in the 6–11 group ($p = .008$, $V = .37$, medium effect), with non-robot showing much higher rates.

The heatmap in Figure 2 illustrates these patterns across age groups. *Collaboration* (L3) was generally higher in non-robot systems, with the widest gap in the 6–11 group, though robot sometimes showed comparable or even higher rates in adolescence. *Active role* (L2) was prevalent across all age groups, with higher percentages in non-robot studies. *Media use* (L1) became widespread from 6–11 onwards, reaching ubiquity in the 11–13 and 13–18 groups for both modalities. *STEM goals* (L4) were somewhat more frequent in non-robot systems, particularly at primary school age and in mixed groups, although overall percentages remained low. *AI support* (L5) appeared relatively stable across most age groups and peaked in mixed-age studies for both robots and non-robots.

These visual observations confirm the aggregate trends in Table 2 and cross-check the age distributions in Table 1.

5 Discussion

5.1 Distribution of artefacts across technology type and age range (RQ1)

The review shows a balanced representation of robot (71/135, 53%) and non-robot phygital artefacts (64/135, 47%). Overall, technology enhanced storytelling studies are concentrated on younger and mixed-age learners, leaving adolescence largely unexplored in HRI research. These age-related gaps are particularly relevant in light of the broader transformation of storytelling toward increasingly mediated and automated practices: if embodied and phygital technologies are to sustain collaborative meaning-making, evidence is needed across developmental stages, not only in early childhood. It should be noted, however, that the non-robot category encompasses a wide variety of phygital solutions, meaning that robots remain consistently represented when compared to any single alternative artefact type. Most studies target mixed-age groups (37% robot, 50% non-robot), followed by the primary school range (6–11 years).

Preschool learners (0–6) are addressed less often, though more by robots (24%) than non-robot ones (11%). Adolescents (11–13 and 13–18) are significantly underrepresented, with only 8% of robot-mediated and 6% of non-robot-based studies. Overall, the literature is skewed toward younger and mixed groups, confirming that storytelling artefacts are primarily designed for early and middle childhood, with limited coverage of older learners.

This finding extends prior scoping reviews in educational robotics and storytelling [111], which similarly reported concentration in early and primary education but provided limited breakdown

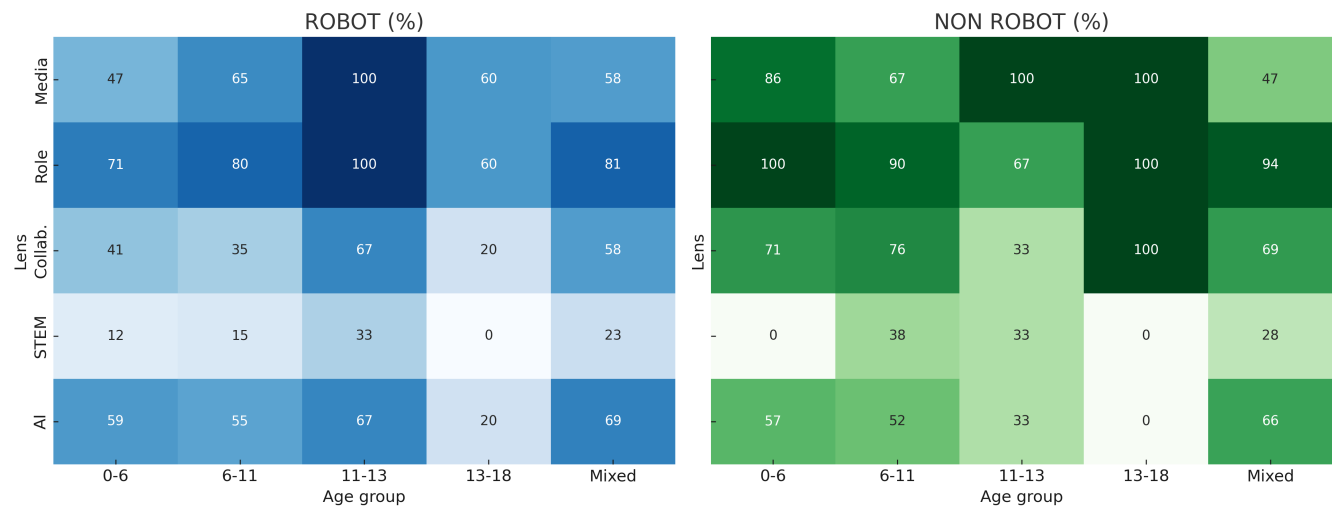


Figure 2: Lens-wise percentages by age range. Left: robots; right: non-robot phygital artefacts. Rows encode age ranges; columns encode lenses (L1–L5). Cell colour indicates the share of studies coded “Yes” (or “Active” for L2) within that age.

across non-overlapping age ranges. By providing explicit denominators per age range, our synthesis clarifies that the literature is strongly weighted toward early childhood and mixed groups, leaving secondary education and adolescence as clear research gaps. The underrepresentation of adolescents (11–18) may reflect curricular priorities, since robots and storytelling are often linked to early literacy and primary STEM education. It may also stem from ethical and motivational challenges in involving teenagers in long-term studies. For HRI design, it implies that current storytelling technologies have been optimised for the developmental needs of younger children, while little is known about how adolescents engage with embodied or phygital storytelling. Future HRI research should explicitly address this gap, as adolescence is a critical stage for identity development and creative self-expression through storytelling.

5.2 Technology-supported phygital artefact comparison (RQ2)

Across storytelling technologies, both robots and non-robot phygital artefacts actively engage children, but in different ways. While non-robot artefacts more consistently foster collaboration and integrate STEM objectives, robots tend to emphasise embodiment, social presence, and more individual or sequential forms of engagement. These differences point to an opportunity for HRI research to better understand how robotic storytelling systems can support collaborative and social learning dynamics.

When comparing robot and non-robot phygital artefacts across the analytic lenses, several patterns emerge. Media (L1) and children’s active participation (L2) are consistently high across both categories (see Figure 2), confirming that storytelling technologies generally promote engagement through multimodal interaction. The active involvement of participants is consistent with the survey by Palioura et al. [111], which reported that 84% of initiatives engaged learners as active contributors to storytelling.

More salient differences emerge in collaboration and STEM learning goals. Non-robot phygital artefacts more often foster synchronous peer collaboration and embed STEM objectives, while robots tend to emphasize novelty, embodiment, and individual or sequential participation. These findings resonate with constructionist perspectives [114], and with prior work on collaboration in educational technology [127]. Robots, while providing social presence, often structured participation sequentially rather than supporting genuine co-construction. In contrast, tangible and phygital artefacts more directly embodied constructionist values by enabling shared ownership of stories. In the 6–11 age range, non-robot phygital artefacts fostered collaboration in 76% of studies, compared to 35% for robots—a statistically significant difference (see Table 2). Similarly, non-robot more often embedded STEM objectives, particularly in primary school and mixed-age contexts, consistent with evidence from tangible and constructionist learning research [57, 108]. Robots, by contrast, were more frequently oriented toward language learning [111] and often structured participation sequentially rather than synchronously.

AI support appeared widespread in both modalities, especially in primary and mixed-age studies, echoing broader trends in child-facing educational technology [67, 159]. Yet most systems relied on rule-based automation or adaptive feedback, with few examples of co-adaptive or generative storytelling. This highlights a gap where transparent and explainable AI could enable children to meaningfully shape the narrative process. While generative AI opens new possibilities for narrative variation, it also raises questions about authorship and agency: when stories are dynamically generated, it may be unclear whether children are co-creating narratives or selecting from system-driven outputs. This tension underscores the importance of transparent and explainable AI in child-facing storytelling systems. Taken together, these results show that while both robots and non-robot phygital artefacts actively engage children, they scaffold narrative practices in complementary ways. Non-robot phygital artefacts emphasise co-construction and domain learning,

Table 3: Design implications (I1–I8) mapped to analytic lenses.

ID	Implication	Lens
I1	Broaden age coverage, especially adolescents; explore screenless mechanisms.	L1
I2	Reduce visual dominance; support embodied, head-up interaction.	L1
I3	Position children as co-creators, not passive listeners.	L2
I4	Diversify roles to broaden agency.	L2
I5	Support synchronous peer co-construction, beyond sequential turns.	L3
I6	Reduce asymmetries in robot vs. toolkit collaboration.	L3
I7	Integrate STEM objectives alongside language and literacy.	L4
I8	Develop transparent, co-adaptive AI beyond simple automation.	L5

whereas robots contribute social presence and embodied novelty. For HRI, this points to the need for designing robots that inherit the collaborative and constructionist strengths of tangible toolkits

5.3 Towards a Complementary Research Agenda for Storytelling Robots

Our review highlights a critical insight: while robots is a promising medium for children’s storytelling, non-robot phygital artefacts more consistently foster collaboration and creative ownership. Robots primarily provide novelty and social presence. This calls for a shift in perspective: rather than positioning robots as substitutes for other phygital artefacts, they should be envisioned as complementary partners within hybrid storytelling ecologies. It is important to note that the design implications proposed in this review are not evaluative claims about system performance, but evidence-informed directions grounded in observed patterns. In particular, the higher frequencies of collaboration (L3) and active participation (L2) in non-robot phygital artefacts highlight opportunities to design storytelling robots that embed similar collaborative and constructionist features, supporting children’s agency rather than asserting modality superiority.

The design implications (I1–I8) derived from the analysis are summarised in Table 3. These implications directly inform a research agenda, *synergistic storytelling in HRI*, which translates the gaps identified in the review into concrete design directions.

Screen-less interaction. Non-robot artefacts often combine tangible interaction with minimal screen use, enabling embodied and inclusive engagement [43, 141]. Robots should draw from this balance, limiting screen dominance and supporting multimodal engagement that enhances rather than constrains children’s immersion (I1, I2).

Hybrid Storytelling Ecosystems. Robots should be integrated alongside tangible and phygital artefacts, acting as facilitators or mediators rather than central protagonists. Future research should investigate hybrid ecosystems where robots enhance narrative richness without restricting children’s agency (I3, I4).

Collaboration-Centric Design. Non-robot phygital artefacts foster peer-to-peer collaboration, whereas robots often structure participation sequentially or promote turn-taking. Future HRI should design for co-construction, with robots supporting negotiation, consensus-building, and shared ownership of stories, scaffolding group dynamics rather than dominating the interaction (I5, I6).

Learning objective. According to the literature, storytelling mediated via robots mainly targets language learning [111], while

non-robotic artefacts display stronger integration of storytelling with STEM skills, aligning with constructionist approaches. Robots should embed similar opportunities, enabling children to program, design, or engineer narrative components, turning storytelling into a gateway for STEM literacy without sacrificing creativity (I7).

Openness and Adaptability. Inspired by the flexibility of non-robot phygital artefacts, robots should support improvisation, personalisation, and repurposing. Designing for openness can move beyond scripted interactions, enabling children to shape both the robot’s role and the story’s evolution. A key direction is advancing AI-driven storytelling robots toward transparent, explainable, and co-adaptive interaction, allowing children to understand and influence the system’s narrative choices (I8).

By drawing inspiration from the collaborative, inclusive, and constructionist strengths of non-robotic artefacts, storytelling robots can move from offering novelty to enabling meaningful and empowering roles in children’s narrative practices. This complementary vision positions robots not as replacements but as synergistic partners that amplify storytelling within HRI.

6 Conclusion, Limitation and Future Directions

This review systematically compared robots and non-robot phygital artefacts in children’s storytelling, analysing 135 studies published between 2014 and 2025. Our synthesis revealed uneven age coverage, with most studies addressing primary school (6 – 11) and mixed-age groups and very limited evidence for adolescents (RQ1). While both modalities engaged learners actively, non-robot phygital artefacts more consistently fostered peer collaboration and creative ownership, whereas robots primarily contributed novelty, embodiment, and social presence, often privileging sequential over co-constructed participation (RQ2). These findings indicate that storytelling robots should be conceived as complementary tools within hybrid narrative ecologies. By incorporating openness, adaptability, and collaboration robots can evolve from offering novelty to enabling meaningful, co-creative learning experiences.

Limitations related to search scope and coding procedures are discussed in Section 3. Beyond these, the evidence base reveals substantive gaps. Inclusivity and accessibility were rarely addressed, leaving open questions about whether screenless interaction can support diverse learners. Although agency was often coded as active, many robot-mediated systems positioned children as listeners rather than co-creators. Finally, the heterogeneity of the robot category, from simple programmable robots to complex social humanoids, calls for finer-grained distinctions in future research.

Future research should systematically integrate collaboration and STEM-oriented goals, and design robots that inherit the openness and adaptability of phygital toolkits, evolving from sources of novelty into synergistic partners in children’s narrative practices.

Data availability. The data supporting this review are accessible at https://github.com/mariaangelapellegrino/StorytellingSLR_RP.

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